

## **Machine Learning for Community Microgrids: Forecasting and Fault Detection in Residential Solar Networks**

*Maryam Sindhu, BSc Student, Faculty of Engineering (Computer Science), University of Ottawa, Ottawa, Ontario, Canada*

Residential neighborhoods with high rooftop solar adoption face persistent challenges including unpredictable demand, limited storage, and the variability of renewable generation. While machine learning (ML) has been widely applied in large-scale utility grids, its use in decentralized, community-level microgrids which are increasingly common in Canadian residential contexts, remains underdeveloped. This project addresses that gap by developing a simulated microgrid testbed and applying complementary ML approaches to improve forecasting and fault detection.

Neural networks were trained on multi-week sequences of household load, solar output, and temperature data to forecast short-term demand. Across walk-forward validation folds, the neural model reduced mean absolute error by 13.27% compared to a linear regression baseline, showing that ML can capture nonlinear interactions such as asymmetric cooling demand and weekend–solar coupling that conventional methods often overlook.

In parallel, decision tree ensembles were developed for storage fault detection, achieving 93.9% anomaly detection accuracy while balancing sensitivity and precision. Feature importance analysis revealed that short-term load lags, solar variability, and time-of-day were the strongest predictors, underscoring that ML does not simply overfit noise but instead extracts physically meaningful drivers of anomalies in microgrids.

Together, these findings provide a proof-of-concept framework where predictive forecasting and diagnostic monitoring work in tandem to enhance grid reliability. By focusing on student housing and suburban neighborhoods as representative case studies, this work highlights how ML-enabled community microgrids can improve local energy resilience and offer a scalable pathway toward sustainable, community-centered power systems.